



Ultra-thin terahertz source

Physicists have developed an extremely thin, large-area semiconductor surface source of terahertz <http://dg.it.in/apr21-49>



Tech vs shark bites

Electronic deterrent devices could reduce the likelihood of shark bites by about 60 per cent <http://dg.it.in/apr21-50>

office that funded the research said, “The results obtained in this work pave the way for developing future generation applications that are of interest to the Department of Defense, such as ultra-dense storage, neuromorphic computing systems, radio-frequency communication systems and more.”

Researchers from the University of Buffalo, in December 2020, announced that they had created an atom thin transistor using MoS₂, and graphene. The transistor consists of a mono atom thin sheet of graphene, next to a mono atom sheet of MoS₂. The transistor is far more efficient than current transistors in the market, and over a decade of use only requires 29 millivolts, as compared to existing versions that operate at 60 millivolts. The transistor has the potential of extending Moore’s Law.

SQUIDS

The US Department of Energy is creating a new facility just to conduct research on new quantum materials, and at the same time develop the specialised tools needed to conduct the studies. This new facility is called the Center for Quantum Sensing and Quantum Materials in Illinois, bringing together experts in the domain from local institutions including SLAC National Accelerator Laboratory, Stanford University and the University of Illinois at Chicago and the University of Illinois at Urbana-Champaign (UIUC). The cutting edge quantum sensing tools being developed include a scanning cubit microscope, a spectroscopy instrument that takes advantage of entangled photons from SLAC’s X-ray free-electron laser, and the Linac Coherent Light Source, which recently got an upgrade. These kinds of photons are paired together and have a strong link that makes one photon change its properties in tandem with any changes in characteristics and properties of its partner. The researchers are also building another material probing and sensing instrument that also takes advantage of entangled photons, called the two-electron Einstein-Podolsky-

Rosen (EPR) spectrometer. Finally a third instrument is being realised for x-ray nonlinear optics.

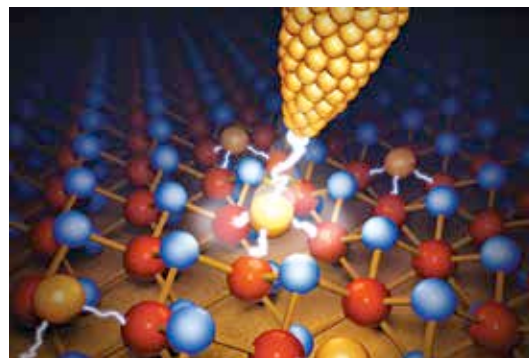
These new methods will allow scientists to understand the underlying physics behind the strange and mysterious ways in which quantum materials behave, allowing them to discover new quantum materials, and eventually create the next generation of probing and sensing tools for the characterisation of quantum materials. Peter Abbamonte, a professor of physics at UIUC and the director of the new institute says, “What is exciting is that this center gives us a chance to create some really new quantum measurement techniques for studying energy-relevant quantum materials. We often get trapped in the cycle of using the same old measurements – not because we don’t need new kinds of information or knowledge, but because developing techniques is expensive and time consuming.”

The problem with understanding these materials is that the matter is squished together in exotic ways with the particles being in an admixture of overlapping states of matter, and entangled with each other. The sensors are being built using a technology known as superconducting quantum interference devices, or SQUIDS, which was actually developed more than half a century ago. SQUIDS are used in the space, defence and nuclear energy sectors for detecting small magnetic fields and superconducting transition edge sensing. The SQUIDS work by putting the sensor in a known quantum state, making it interact with the quantum material of interest, and measuring the precise changes to the state of the quantum system because of the interaction. The scanning cubit microscope is expected to have a number of cubit probes on its sensor tip, which can be a single hydrogen atom whose electron is simultaneously in multiple states. By entangling the qubit sensor interacts

with the quantum states within the material being probed.

TWISTRONICS

Researchers discovered that by slightly twisting some 2d quantum nanomaterials they can acquire interesting new electrical properties, including unconventional superconducting properties, kicking off a field of research known as twistrionics. For example, a material may completely lose its electrical resistance at low temperatures. The conventional approach for creating twisted 2d materials is to stack them, then precisely control the amount of mechanical twist. When these materials are grown the interaction between them is weak, so they cannot be twisted at that time itself. The researchers used



World’s smallest memory storage unit

non-Euclidean geometry on the crystal structures to figure out a new approach for twisting the two sheets of material as they grow. The materials used were transition metal dichalcogenides, but the principle applies to other nanomaterials as well. The team took advantage of a type of imperfection in crystal growth called screw dislocations, which in 2D materials cause subsequent layers of the structure to twist on their own, like a spiral staircase. The team introduced a tiny imperfection, a nanoparticle on the substrate on which the crystal is grown, creating a curved foundation for the growth, and twisting the sheets of the materials in a predictable way, creating a cone shaped base. The direct synthesis of twisted 2D quantum materials simplifies future twistrionics research. 